



PROFILE

Multimodal AI Engineer and Researcher working at the intersection of language, vision, and human cognition. 5+ years of experience spanning industry data science, applied ML research, and computational neuroscience.

Specialized in multimodal deep learning architectures combining transformer language models with vision encoders.

Background in end-to-end ML engineering: from model design and fine-tuning to production deployment with FastAPI, Docker, and Azure.

Currently extending this into generative AI applications — building retrieval-augmented generation systems and applying the same production discipline to bridge research-grade models with real-world adoption.

CONTACT

✉ diegoblassio9@gmail.com

📍 Valencia, Spain

🌐 dblassio.github.io

🔗 [Dblassio](#)

DIEGO BLASSIO

Multimodal AI Engineer | NLP · Computer Vision · LLMs · NeuroAI

WORK EXPERIENCE

Marsh McLennan - New York Team

2022-2024

Data Scientist

- **Built and deployed end-to-end machine learning models**, including anomaly detection systems combining supervised and unsupervised techniques, with interpretability through SHAP values and large language models.
- **Optimized model performance across key metrics** (F1-Score, Recall, Precision, Accuracy, Silhouette Score, False Positive Rate), achieving measurable improvements in anomaly detection reliability.
- **Designed and implemented scalable infrastructures with FastAPI and Docker**, ensuring deployment on OSS2 and Azure, while effectively communicating complex analyses to non-technical stakeholders for strategic decision-making.

Getin - Mexico Team

2019-2022

Data Analyst

- **Led data-driven analysis in the retail sector**, automating reporting processes with R and Visual Basic to deliver 500+ monthly/daily reports for senior executives and decision-makers.
- **Produced and published statistical insights** on retail industry trends and performance metrics in reputable newspapers, translating complex analyses into accessible insights for non-technical audiences.
- **Designed interactive dashboards and visualizations** with Tableau, supporting ad-hoc analyses of seasonal patterns and market trends that drove actionable strategic decisions.

Metlife & Metrofinanciera

2016 - 2019

Finance Trainee

EDUCATION

M.Sc. in Computational Neuroscience

2024-2026

Université Côte d'Azur, France

Focus: Computational neuroscience, brain dynamics, cognitive modeling, and applied deep learning

Artificial Intelligence Professional Program

2023

Stanford University, USA

Courses: Deep Multi-Task & Meta Learning, Machine Learning with Graphs, Natural Language Processing with Deep Learning.

Data Science & Machine Learning Program

2021

ITAM, Mexico

Topics: Supervised/Unsupervised ML, Deep Learning (ANN, CNN, LSTM, RNN).

B.Sc. in Actuarial Science

2019

ITAM, Mexico

Thesis: ARIMAX time-series modeling.

LANGUAGES

- Spanish (Native)
- English (C2)
- French (B1/B2)

CERTIFICATIONS

- AWS Certified AI Practitioner (In Progress) (Exam scheduled: August 2026)

OUTSIDE THE LAB

- 3x Triathlons
- 4x Half Marathons
- 1x Marathon
- 3x L'Etape Road Cycling
- 2x High Mountain Summits

APPLIED RESEARCH PROJECTS

RAG Pipeline — arXiv Retrieval System (In Development) 2026

- Building a Retrieval-Augmented Generation pipeline from arXiv abstracts: tokenizer-aware chunking, dense embeddings (BAAI/bge-small-en-v1.5), and persistent vector indexing (ChromaDB, cosine similarity search).
- Extending the system toward multimodal retrieval (text + image), applying adaptive modality-weighting principles from prior published research (CLEF 2026 gated fusion architecture) to retrieval design.
- Public repository: [GitHub](#)

UPV - PRHLT Team (Valencia, Spain)

2026

Human-Centered Multimodal AI (Under Prof. Paolo Rosso)

- Designing multimodal architectures that fuse textual, visual, and physiological signals (EEG, eye-tracking, heart rate) for tasks involving implicit human judgment.
- Applying human-centered AI principles to sexism detection in social media, using involuntary biological responses as supervisory signal to improve model alignment with real human perception.
- Exploring cross-modal learning strategies for content classification and semantic categorization across images, text, and video.
- Public code repository: [GitHub](#).

INRIA — MARIANNE Team (Nice, France)

2025-2026

Multimodal NLP for Argument Mining (Under Prof. Serena Villata)

- Co-authored "Multimodal Stance Detection in Controversial Tweets", COMMA 2026 (International Conference on Computational Models of Argument, Barcelona)
- Developed and benchmarked multimodal architectures combining DeBERTa-v3 with vision encoders (CLIP, BLIP, ALIGN, ViT, ResNet), evaluating five fusion strategies: mean pooling, weighted mean, gated fusion, concatenation, and projected concatenation.
- Re-annotated the ImageArg dataset with two independent annotators (Fleiss' $\kappa = 0.865$), correcting labelling errors and achieving +8.7 F1 points improvement across all model types.
- Demonstrated that images carry independent stance-relevant information (CLIP vision-only: $F1 = 0.725$) providing computational evidence for visual argumentation theory.
- Public code repository [GitHub](#).

PUBLICATIONS

- Blassio, D., Dore, D., Cabrio, E., & Villata, S. "Multimodal Stance Detection in Controversial Tweets." Proceedings of COMMA 2026 – 11th International Conference on Computational Models of Argument, Barcelona, 2026.
- Blassio, D. "Gated Multimodal Fusion with Neurophysiological Signals for Sexism Detection in Memes." Working Notes of the EXIST Lab at CLEF 2026
- M.Sc. Thesis: "Gated EEG Transformers for Multimodal Sexism Detection in Memes." Université Côte d'Azur, 2026.
Advisors: Prof. Paolo Rosso, Dr. Marco Siino.

KEY SKILLS

- **Programming & Data:** Python (pandas, numpy, scipy), R, SQL
- **ML & Deep Learning:** PyTorch, Hugging Face, scikit-learn, Transformers (DeBERTa, RoBERTa, CLIP, BLIP, ViT), multimodal fusion strategies, fine-tuning (PEFT/LoRA), anomaly detection, time-series forecasting, SHAP
- **Deployment & MLOps:** FastAPI, Docker, DVC, Git/GitHub, CI/CD, Azure, automated retraining pipelines
- **Research & Analysis:** experimental design, ablation studies, inter-annotator agreement, fMRI/EEG signal processing, statistical modeling
- **Visualization:** Tableau, matplotlib, Seaborn, scientific visualizations